Q.1 (A) 28,600

initial population = 20,000

first year increase:

$$20000 \times \frac{110}{100} = 22000$$

second year increase:

$$22000 \times \frac{120}{100} = 26400$$

final answer is 26,400.

Q.2 (B) 12,000

winner's votes = 58%

Loser's votes: 100 - 58 = 42%

Difference: 58 - 42 = 16%

16% or total votes = 1.920.

$$\text{total} = \frac{1920 \times 100}{16}$$

$$= 12000$$

final answer : 12,000 votes.

Q.3 (C) 23.46%

Cost price of 10 pens = ₹9
Cost price of 1 pen: CP = 9/10 = ₹0.90
Selling price of 1 pen: = ₹10
Selling price of 1 pen: SP = 10/9 = 1.1111
Profit percentage: = (SP - CP/CP) × 100

$$= \frac{1.1111 - 0.90}{0.90} \times 100$$

$$= 23.46\%$$

Final answer: 23.46%.

Q.4 (B) 12.5%

Assume CP = ₹100

Marked price: MP = 100 + 25 = ₹125

Discount: 125 × 10/100 = ₹12.50

Selling price: 125 - 12.50 = ₹112.50

Profit:

112.50 - 100 = ₹12.50

final answer: 12.5%

Q.5 ~~(A) 10%~~ (B) 11 1/9%

Difference between amounts ₹400 - ₹200 = ₹1200
this is the interest for 2 years
Interest for 1 year: 1200 ÷ 2 = ₹600
principal: 7200 - (600 × 3) = ₹5400
simple interest formula:

$$R = SI \times 100 / P \times T$$

$$R = 600 \times 100 / 5400$$

$$R = 11.11\%$$

Final answer is $11\frac{1}{9}$

Q.6 (B) ₹1,640

Since interest is compounded half-yearly
Rate per half-year: 10 ÷ 2 = 5%
number of periods = 2

$$A = 16000 (1 + 5/100)^2$$

$$= 16000 (1.05)^2$$

$$= 16000 \times 1.1025$$

$$= A = ₹17640$$

compound interest

$$CI = 17640 - 16000 = ₹1640$$

final answer : 1,640.

Q.7 (B) 15 years

Given · 5 years = 2 times
10 years = 4 times
15 years = 8 times

Because $8 = 2^3$

$\therefore$ $5 \times 3 = 15$

Final answer : 15 years

Q.8 (B) 7.2 days

A's one day work 1/12

B's one-day work 1/18

together 1/12 + 1/18

LCM = 36.

$= \frac{3+2}{36} = \frac{5}{36}$

total time

$= \frac{36}{5} = 7.2$

Final answer : 7.2 days

Q.9 (C) 30 days

together one-day work: 1/10

A's one-day work: 1/15

B's one day work 1/10 - 1/15

= 3-2/30 = 1/30

Final answer: 30 day

Q.10 (D) 60 minutes

A's filling rate 1/20

B's emptying rate 1/30

Net filling rate:

$$\frac{1}{20} - \frac{1}{30}$$

$$= \frac{3-2}{60}$$

$$= \frac{1}{60}$$

Final answer: 60 minutes.

Q.11 (A) 18 Km/h

For equal distances, average speed is:

$$\text{Average} = \frac{3}{1/10 + 1/20 + 1/60}$$

LCM = 60

$$= \frac{3}{6+3+1/60}$$

$$= \frac{3 \times 60}{10}$$

$$= 18 \text{ km/h}$$

Final answer: 18 km/h.

Q.12 (B) 40 minutes

Let: usual time = x minutes.
When speed becomes 4/5, time becomes 5/4
Extra time

$$\frac{5}{4}x - x = 10$$

$$\frac{1}{4}x = 10$$

$$x = 10 \times 4 = 40$$

Final answer: 40 mintes.

Q13 (B) 72 Km/h

time = 12 seconds

Distance = 240 m

$$\text{speed} = \frac{240}{12} = 20 \text{ m/s}$$

Convert into Km/h:

$$20 \times \frac{18}{5} = 72$$

Final answer: 72 Km/h.

Q.14 (C) 320 m

Convert

$$90 \times \frac{5}{18} = 25 \text{ m/s}$$

total distance covered:

$$25 \times 20 = 500 \text{ m}$$

Platform Length :

$$= 500 - 180$$

$$= 320 \text{ m}$$

final answer : 320 metes.

Q.15 (B) 12 seconds

total length 120 + 180 = 300 m

Relative speed (opposite directions):

50 + 40 = 90 km/h

Convert:

90 × 5/18 = 25 m/s

time: = 300/25 = 12

Final answer: 12 seconds.

Q.16 (B) 3 km/h

$$\text{Stream speed} = \frac{\text{Downstream} - \text{upstream}}{2}$$

$$= \frac{18 - 12}{2}$$

$$= \frac{6}{2} = 3$$

Final answer: 3 km/h.

Q.17 (B) 6:4:3

Let : $2A = 3B = 4C = K$

then : $A = K/2$, $B = K/3$, $C = K/4$

$\therefore$ $A : B : C = 1/2 : 1/3 : 1/4$

multiply by Lcm 12 : $= 6 : 4 : 3$

Final answer : 6:4:3

Q.18 (C) ₹12000

A investment $40000 \times 12 = 480000$

B investment $60000 \times 6 = 360000$

Ratio $480000 : 360000 = 4:3$

total parts = 7.

A's share $= \frac{4}{7} \times 21000$

$= ₹12000$

final answer : ₹12,000.

Q.19. (C) 36

Sum of 5 numbers

$5 \times 24 = 120$

Sum of remaining 4 numbers: $4 \times 21 = 84$

Removed numbers: $120 - 84 = 36$

Final answer: 36.

Q.20 (B) 37

Let the average after 16 innings $= x$.
new average after 17 innings $= x + 3$
Using the total runs:

$$16x + 85 = 17(x + 3)$$

$$16x + 85 = 17x + 51$$

$$85 - 51 = 17x - 16x$$

$$x = 34$$

new average:

$$34 + 3 = 37$$

Final answer: 37 runs.

Q.21. (B) 10 years

Let daughter's present age = x.
mother's present age : 50 - x

Five years ago:
- Daughter = x - 5
- Mother = 50 - x - 5 = 45 - x

According to the question:

$$45 - x = 7(x - 5)$$
$$45 - x = 7x - 35$$
$$80 = 8x$$
$$x = 10$$

Final answer : 10 years.

Q.22 (C) 15 years

Let son's present age = x.
Father's present age = 3x
After 15 years:

$$3x + 15 = 2(x + 15)$$
$$3x + 15 = 2x + 30$$
$$3x - 2x = 30 - 15$$
$$x = 15$$

Final answer: 15 years.

Q.23. (B) 2 : 1

Selling Price = ₹ 230.
Profit = 15%
First find the cost price of the mixture:

$$CP = 230 \times 100 / 115$$

$$CP = ₹200$$

Using the allegation method:

$$\text{Ratio} = 240 - 200 / 200 - 180$$

$$= 40/20 \quad = 2:1$$

Final answer: 2:1.

Q.24. (A) 4 litres.
total mixture = 80 litres.

Milk : $7/10 \times 80 = 56\ L$

water: $3/10 \times 80 = 24L$
required ratio = 2:1.
milk remains 56 litres.

Required water: $56 \div 2 = 28L$

Additional water: $28 - 24 = 4L$

Final answer : 4 Litres.

Q.25 (C) 600

Lets the numbers be x and y. $x+y=50$

$x+y=10$

Adding both equations: $2x=60$

$x=30$

$\therefore \quad y=50-30$

$=20$

Product: $30\times20=600$

Final answers: 600.

Q.26 (B) 72

Formula:

$$\text{First} \times \text{second} = \text{HCF} \times \text{LCM}$$

Let the second number $=x$.

$$60\times x=12\times360$$

$$60x=4320$$

$$x=\frac{4320}{60}$$

$$x=72$$

Final answer: 72.

Q.27 (C) 94.

First find the LCM of 6, 9, and 15.

$6 = 2 \times 3$

$9 = 3^2$

$15 = 3 \times 5$

LCM

$2 \times 3^2 \times 5 = 90$

Required number:

$90 + 4 = 94.$

Final answer: 94.

Q.28. (B) 130°

Where

- H = 3
- m = 40

Substitute $= [30(3) - 5.5(40)]$

$= |90 - 220|$

$= 130°$

Final answer.

Q.29. (C) thursday.

So 2025 is not a leap year.
number of days = 365.

$365 \div 7 = 52$ weeks + 1 day

∴ the next year starts one day ahead.

wednesday + 1 day = thrusday.
final answer: thrusday.

Q.30 (B) 19 metres.

∴ A:B = 100:90.
when B runs 100m, C runs 90m.

B:C = 100:90

when A runs 100m, B runs 90m.
C's distance:

$$= \frac{90}{100} \times 90$$

$$= 81m$$

H.:, A beats C by :

$$100 - 81 = 19m$$

Final answer: 19 metres.